

Zilong Li

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SPECIALIZATION

Aerospace Engineering with a focus on
Computational Fluid Dynamics (CFD), Machine Learning and Reduced Order Modeling.

EDUCATION

Iowa State University (ISU) 08/2021-Present
Ph.D. in Aerospace Engineering Major GPA: 4.0/4.0; Overall GPA: 4.0/4.0

The Pennsylvania State University, University Park (PSU) 08/2017-12/2019
M.S. in Civil Engineering Major GPA: 3.96/4.0; Overall GPA: 3.92/4.0
M.S. minor in Computational Science GPA: 4.0/4.0

Shanghai Jiao Tong University (SJTU) 09/2012-03/2015
M.S. in Fluid Mechanics Major GPA: 87/100; Overall GPA: 85/100

Wuhan University of Science and Technology (WUST) 09/2008-06/2012
B.S. in Engineering Mechanics Major GPA: 89/100; Overall GPA: 87/100 Rank: 3/40

RESEARCH EXPERIENCE

08/2021-Present Department of Aerospace Engineering, ISU
Research Assistant in *Aerospace Engineering*

- Lead a project on Reduced Order Modeling (ROM)
- Implemented and validated reduced order models in OpenFOAM
- Machine Learning enabled turbulence models for unsteady aerodynamic flows

08/2017-12/2019 Department of Civil and Environmental Engineering, PSU
Research Assistant in *Civil Engineering*

- Collaborated on a project on sediment transport
- Implementation of Smoothed Particle Hydrodynamics (SPH) method in DualSPHysics for granular flow
- High-fidelity numerical simulation and analysis of turbulence

06/2015-12/2015 Estuarine and Coastal Science Research Center, Shanghai
Research Assistant in *Computational Fluid Dynamics*

- Lead a field investigation of flow around groins in Hangzhou Bay
- Completed experimental investigation of flow around artificial groins and numerical simulation of the flow field
- Experience with High Performance Computing (HPC)

03/2013-03/2015 Department of Engineering Mechanics, SJTU
Research Assistant in *Computational Fluid Dynamics*

- Lead a project on flow characteristics around spur dikes in an open channel
- Rich experience in computer programming, algorithms, and numerical analysis

- Enabled large scale simulations on multiple CPUs via OpenMP
- 06/2013-03/2014 Ministry of Education (MOE) Key Laboratory of Hydrodynamics, SJTU
Research Assistant in *Particle Image Velocimetry (PIV) Experiment*
 - Independently conducted a PIV experiment to measure the flow field
 - Assisted in a project on flow measurements around bridge piers
- 09/2011-06/2012 Department of Engineering Mechanics, WUST
Research Assistant in *Engineering Mechanics*
 - Assisted in a project on carbon fiber reinforced concrete design
 - Gained the solid knowledge of Finite Element Method (FEM)

PUBLICATIONS

1. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. Accepted by the *AIAA Journal*. Doi: 10.48550/arXiv.2511.18276
2. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Accurate Predictions of Time Resolved Unsteady Flows Over Airfoils. *AIAA SciTech Forum*, 2025. Doi: 10.2514/6.2025-2049
3. **Z. Li**, P. He. Accelerating Unsteady Aerodynamic Simulations Using Predictive Reduced order Modeling. *Aerospace Science and Technology*, 2023. Doi: 10.1016/j.ast.2023.108412
4. **Z. Li**, P. He. Airfoil Unsteady Aerodynamic Analysis Using a Galerkin Reduced order Modeling Approach. *AIAA SciTech Forum*, 2022. Doi: 10.2514/6.2022-0080
5. **Z. Li**, J. Kou, & J. Zhang. Numerical simulations of flows around a single spur dike in an open channel. *Chinese Journal of Computational Mechanics*, 2016. Doi: 10.7511/jslx201602016

PAPERS IN PREPARATION

- **Z. Li**, L. Fang, A. Sharma, and P. He. “Augment RANS Turbulence Models for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall Using High Fidelity DES/LES data.” (In preparation).

TEACHING EXPERIENCE

- 01/2022-Present Department of Aerospace Engineering, ISU
Teaching Assistant in *Aerospace Systems Integration, Advanced Flight Structures, Design of Aerospace Systems, Modern Design Methodology with Aerospace Applications*

COMPUTER SKILLS

Programming Languages: Strong proficiency in C/C++, Python, Fortran, MATLAB
 CFD Packages: Extensive experience with OpenFOAM
 High Performance Computing Platform: TACC Stampede3, NOVA Cluster
 Operating Systems: Expertise in Linux (Ubuntu, Fedora)

AWARDS & HONORS

- 2025 Research Excellence Award in the College of Engineering, ISU
- 2018 H. Marcus Dean's Chair Scholarship in the College of Engineering, PSU
- 2018 Harry G. Miller Scholarship in the College of Engineering, PSU
- 2017 University Graduate Fellowship (UGF) in the College of Engineering, PSU

- 2014 2nd Prize Scholarship (Top 15%), SJTU
- 2013 1st Prize in National Graduate Mathematical Modeling Contest (Top 3%)
- 2012 Outstanding Graduate Student Award (Top 10%), WUST
- 2011 3rd Prize in National Undergraduate Mathematical Contest in Modeling (Twice)
- 2010 National Encouragement Scholarship (Top 5%), WUST
- 2010 3rd Prize in National College Student Basic Mechanics Experiment Contest (Top 3%)
- 2010 “Excellent” Prize in National Pei-Yuan Chou Mechanics Competition

EXTRACURRICULAR ACTIVITIES

- 2013-2014 **Assistant**, Employment Guidance Center, SJTU
- 2010-2012 **Vice-Chairman**, Mathematical Modeling Association, WUST
- 2009-2012 **Core Member**, Wuhan University of Science and Technology Table Tennis Club

REFERENCES

Dr. Ping He

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Dr. Anupam Sharma

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Iowa State University, Ames
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